Investigating Specific Heat Capacity

Exam code: 0625 & 0972

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Investigating specific heat capacity

Extended tier only

Aim of the experiment

- The aim of the experiment is to determine the specific heat capacity of a substance, by linking the decrease of one energy store (or work done) to the increase in temperature and subsequent increase in thermal energy stored

Variables

- **Independent variable** = Time, t
- **Dependent variable** = Temperature, θ
- Control variables:
  - Material of the block
  - Current supplied, I
  - Potential difference supplied, V

Equipment

Equipment list

Equipment	Purpose
Thermometer	To measure the temperature change of the solid / the water
Solid block of aluminium	To investigate temperature changes
Beaker of water (400 ml)	To investigate temperature changes
Immersion heater	To heat the solid / the water
Voltmeter	To measure the voltage across the immersion heater

Stuck on something?
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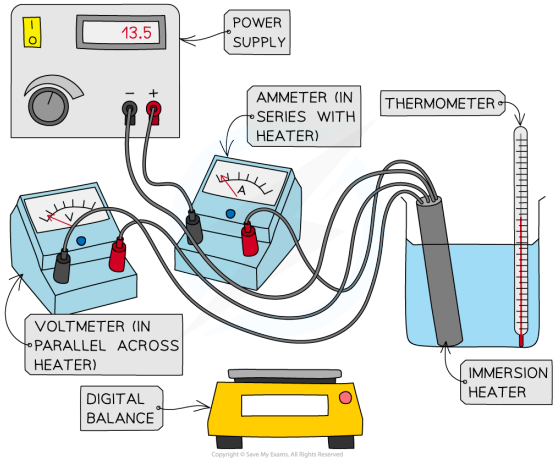
Got it

Power supply	To supply power to the immersion heater
Digital balance	To measure the mass of the solid / the water
Stopwatch	To time the heating of the solid / the water

- **Resolution** of measuring equipment:
  - Thermometer = 0.1 °C
  - Voltmeter = 0.1 V
  - Ammeter = 0.1 A
  - Stopwatch = 0.01 s
  - Digital balance = 0.1 g

Method

The experiment set up for the specific heat capacity practical



Apparatus for heating water and measuring energy supplied

1. Place the beaker on the digital balance and press 'zero'
2. Add approximately 250 ml of water and record the mass of the water using the digital balance
3. Place the immersion heater and thermometer in the water
4. Connect up the circuit as shown in the diagram, with the ammeter in series with the power supply and immersion heater, and the voltmeter in parallel with the immersion heater
5. Record the initial temperature of the water at time 0 s
6. Turn on the power supply, set it at approximately 10 V, and start the stopwatch
7. Record the voltage from the voltmeter and the current from the ammeter
8. Continue to record the temperature, voltage and current every 60 seconds for 10 minutes
9. Repeat steps 2–8, replacing the beaker of water for the solid block of aluminium and starting with recording its mass using the digital balance

specific heat capacity practical

ENERGY SUPPLIED = VOLTAGE × CURRENT × TIME

TAKE EACH NEW TEMPERATURE AND SUBTRACT THE INITIAL TEMPERATURE

TIME / s	VOLTAGE / V	CURRENT / A	TEMPERATURE / °C	ENERGY SUPPLIED / J	TEMPERATURE CHANGE / °C
0			INITIAL TEMPERATURE	0	0
60					
120					
180					
240					
300					
360					
420					
480					
540					
600					

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An example of a suitable results table for the specific heat capacity experiment looks like this. Energy supplies = voltage x current x time.

Analysis of results

- The thermal energy supplied to the block can be calculated using the equation:

$$E = IVt$$

- Where:
  - E = thermal energy, in joules (J)
  - I = current, in amperes (A)
  - V = potential difference, in volts (V)
  - t = time, in seconds (s)

- The change in thermal energy is defined by the equation:

$$\Delta E = mc\Delta\theta$$

- Where:
  - ΔE = change in thermal energy, in joules (J)
  - m = mass, in kilograms (kg)
  - c = specific heat capacity, in joules per kilogram per degree Celsius (J/kg °C)
  - $\Delta\theta$ = change in temperature, in degrees Celsius (°C)

- Rearranging for the specific heat capacity, c:

$$c = \frac{\Delta E}{m\Delta\theta}$$

- To calculate Δθ:

Δθ = final temperature – initial temperature

- To calculate ΔE:

$$\Delta E = IV\Delta t$$

- Where:
  - I = average current, in amperes (A)
  - V = average potential difference (V)
  - Δt = time spent heating, in seconds (s)

- These values are then substituted into the specific heat capacity equation to calculate the specific heat capacity of the aluminium block

Systematic Errors:

- Ensure the digital balance is set to zero before taking measurements of mass
- Some water may be lost to the surroundings by evaporation. Calculate an average mass of water (using the mass before the experiment and the mass after) to account for this
- Remember to only take gradients on the straight-line region
  - Before this point the energy supplied is being used to heat the immersion heater itself

Random Errors:

- Stir the water constantly whilst heating it to ensure the temperature measured is the temperature throughout the fluid
- When the current or voltage values appear to be changing between two values next to one another then be consistent in choosing the higher value

Safety considerations for the specific heat capacity practical

- The immersion heater will get very **hot**
  - Make sure not to touch it, and have a heatproof mat ready to place it on
- Make sure that the immersion heater is connected to a **Direct Current** supply
- The beaker may become **unstable** with an immersion heater and thermometer resting in it
  - If you feel this is the case then use a clamp stand to hold both
- Wear goggles while heating water
- Make sure to stand up during the whole experiment, to react quickly to any spills

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